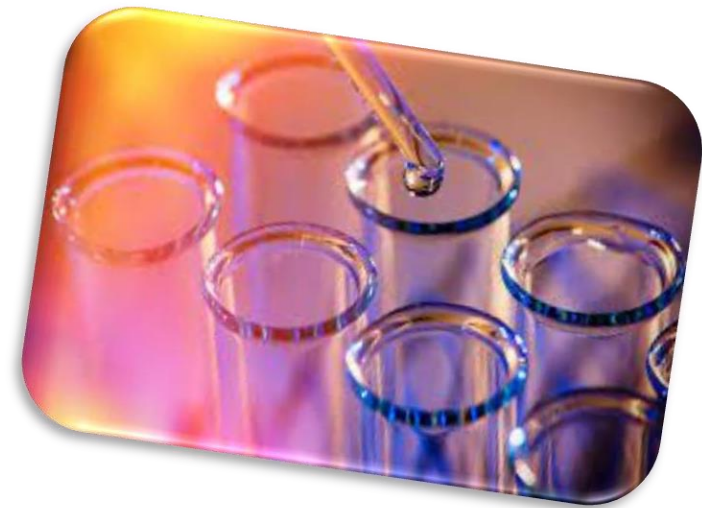




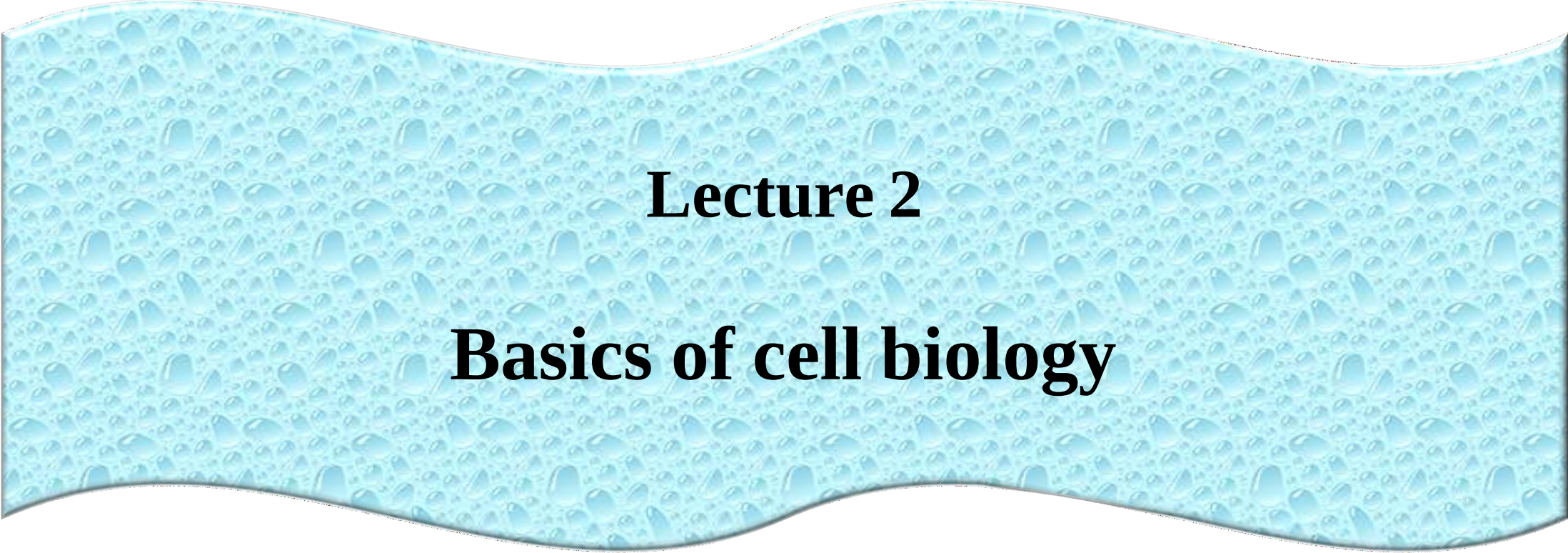
Arab Academy
for Science, Technology & Maritime Transport

Biochemistry EBA1300



Dr. Mohamed R. Eletmany

PhD in Fiber Science and Polymer Chemistry
NC State Wilson College of Textiles, NC State
University, North Carolina, USA

A horizontal banner with a wavy, undulating top and bottom edge. The background is a light blue color, and it is covered with a dense pattern of small, realistic-looking water droplets of various sizes. The droplets have highlights and shadows, giving them a three-dimensional appearance.

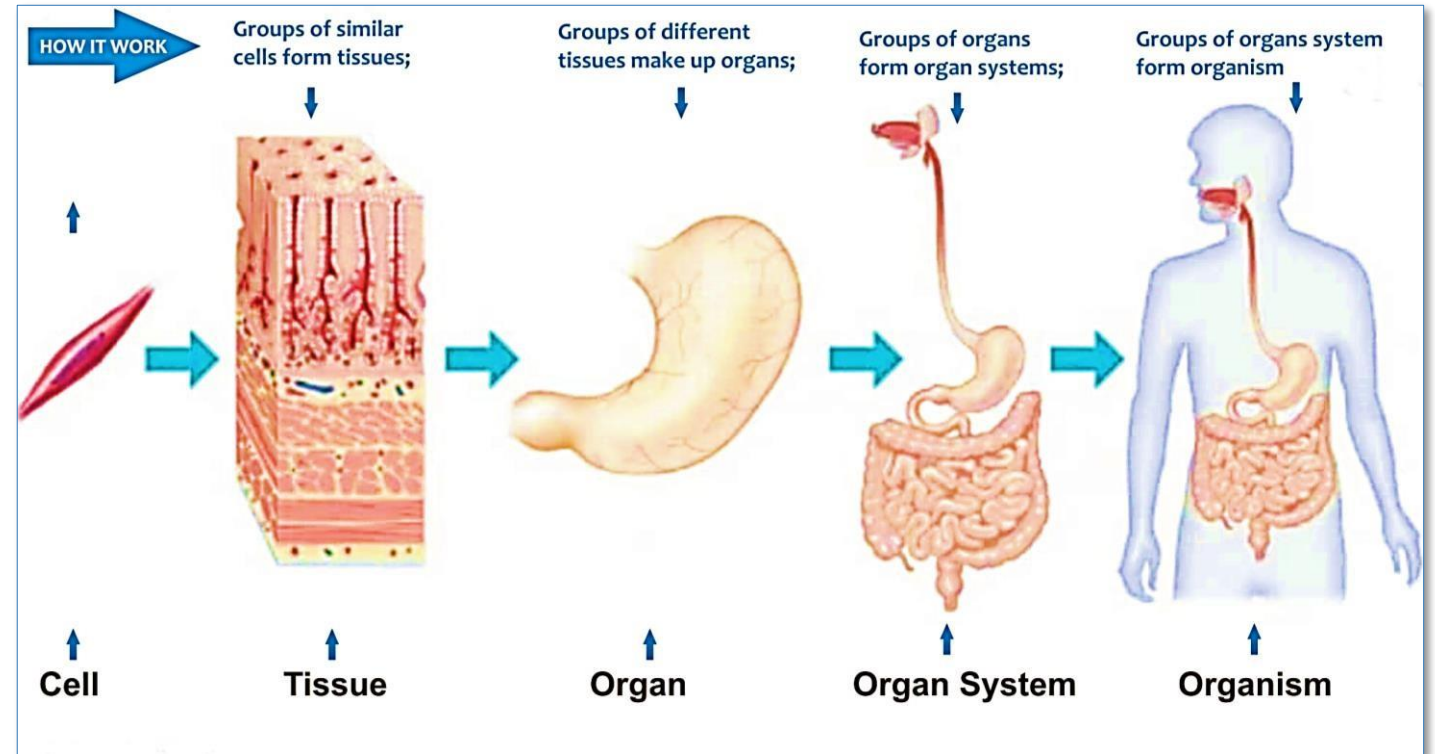
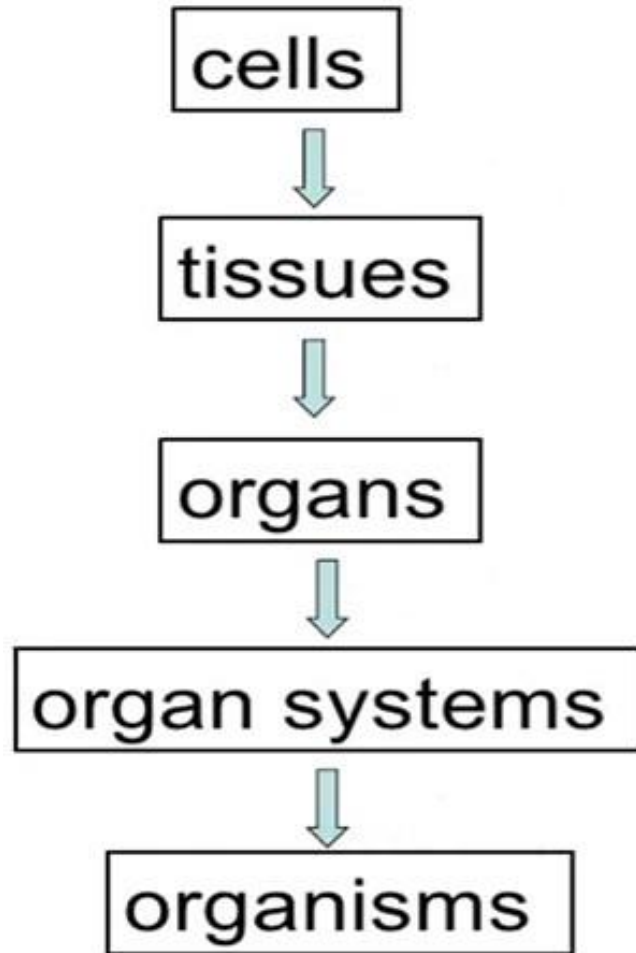
Lecture 2

Basics of cell biology

What is Cell Biology ?

- **Cell biology:** is the study of structure, function, growth, reproduction and genetics of the **cell**.
- **The Cell** is the smallest basic unit forming the living organism (the building block of an organism).
- Cell is a Latin word which means a **small room**.
- Some organisms have only one cell (bacteria) , while others are composed of a huge numbers of cells (human).

Levels of organisation



Discovery of the cell

Scientists contributed to the cell discovery

Hans and Zacharias Jansen



Robert Hooke



Antony van Leeuwenhoek



Robert Brown



Discovery of the cell

Scientists contributed to the cell discovery

Matthias Schleiden

A light blue downward-pointing arrow indicating a sequence or flow from Matthias Schleiden to Theodor Schwann.

Theodor Schwann

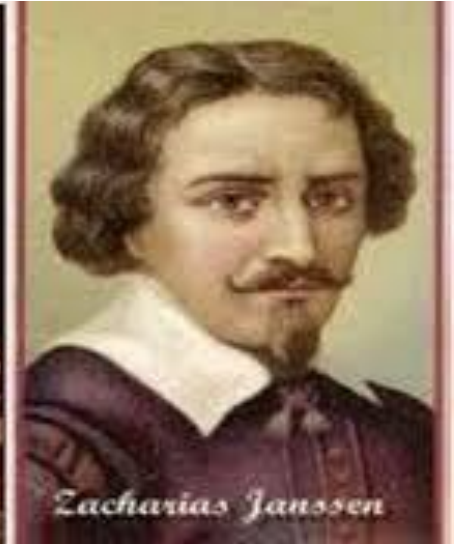
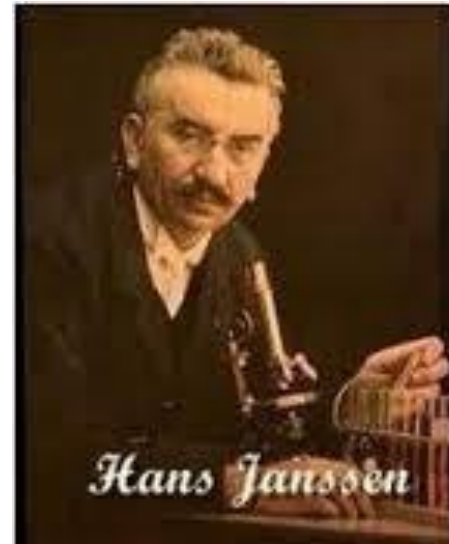
A light blue downward-pointing arrow indicating a sequence or flow from Theodor Schwann to Rudolph Virchow.

Rudolph Virchow

Hans and Zacharias Jansen

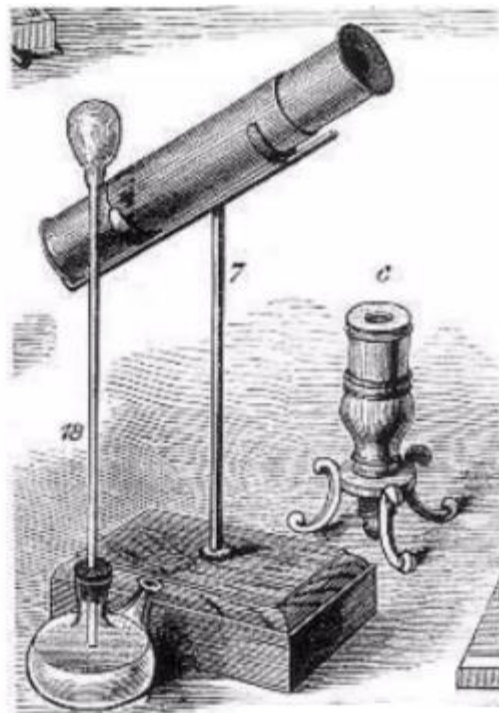
1595

Role
Production of
lenses (compound
microscope)





The First
Compound
Microscope
(circa 1595)





Robert Hooke

- A British biologist and microscopist.
- The word cell was first used by Robert Hooke.
- Hooke looked at thin slices of cork under a microscope, and described what he saw as a honeycomb as it was made up of many tiny units.



Oak tree

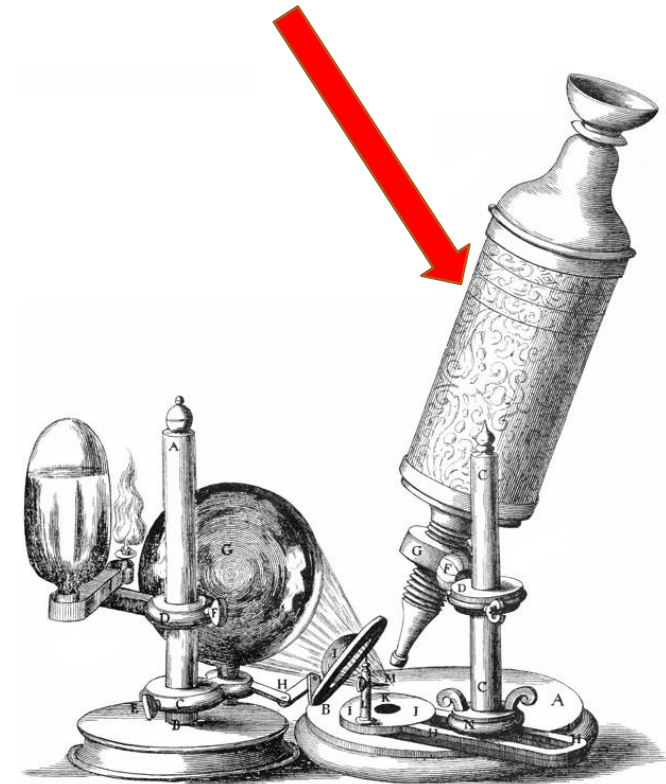


Cork



Discovery of the cell

Hooke's drawing of cork is shown in this Figure, using Hooke's microscope.





Antony van Leeuwenhoek

- During the 1670s, the Dutch Antony van Leeuwenhoek, used microscope to observe many microbes and body cells.
- Leeuwenhoek microscope was able to magnify much more clearly than other microscopes (magnification power over 200 times compared to other microscopes).
- He discovered blood cells and was the first person to see living sperm cells.

Robert Brown

1833

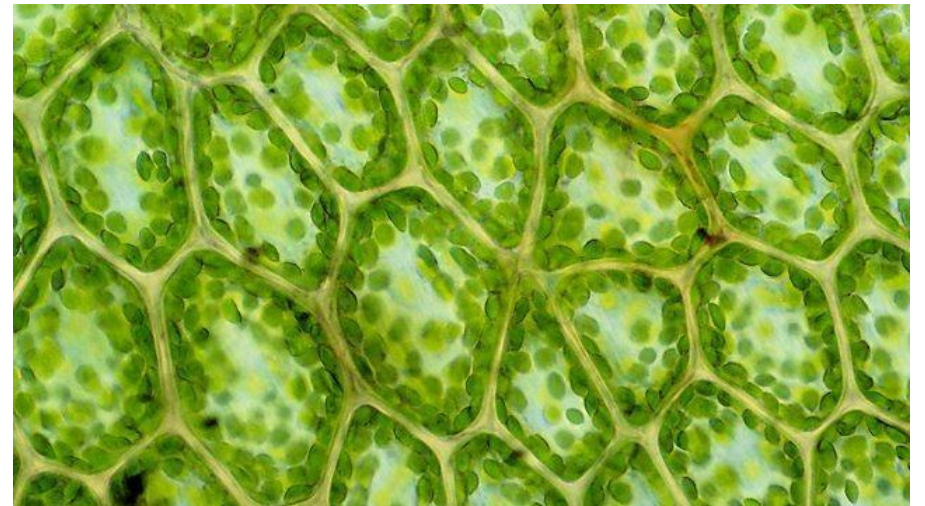
He was a an
English botanist
discovered the
nucleus in Orchid
plant cell



Matthias Schleiden

1838

German botanist
discovered that
cells are the
basic units of
plants

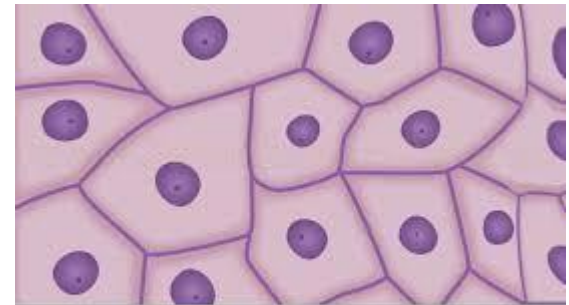


Theodor Schwann

1839

German zoologist
discovered that all
animal tissues are
composed of cells

Started cell theory



Animal Tissues

Rudolph Virchow

1858

Completed cell
theory



Cell Theory

The modern cell theory states that:

- All organisms are made up of one or more cell(s).
- All the life functions of an organism occur within cells.
- All cells come from preexisting cells.

Prokaryotes vs. Eukaryotes

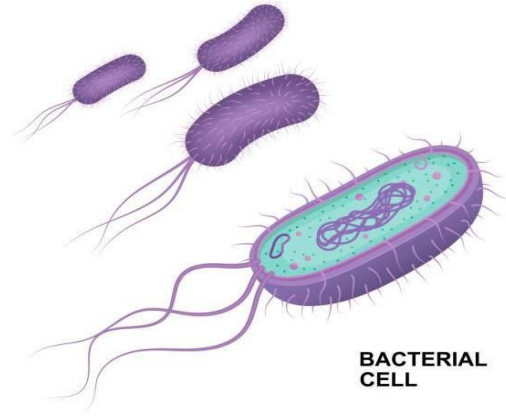
There are two main types
of cells

```
graph TD; A[There are two main types of cells] --> B[Prokaryotic cells (have false nucleus)]; A --> C[Eukaryotic cells (have true nucleus)];
```

Prokaryotic cells
(have **false** nucleus)

Eukaryotic cells
(have **true** nucleus)

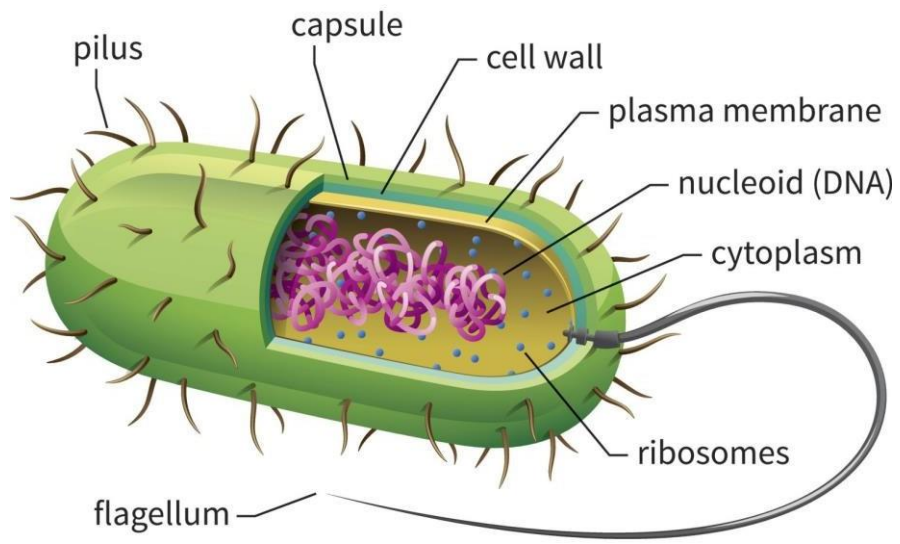
Prokaryotes



Blue-green algae

Eukaryotes





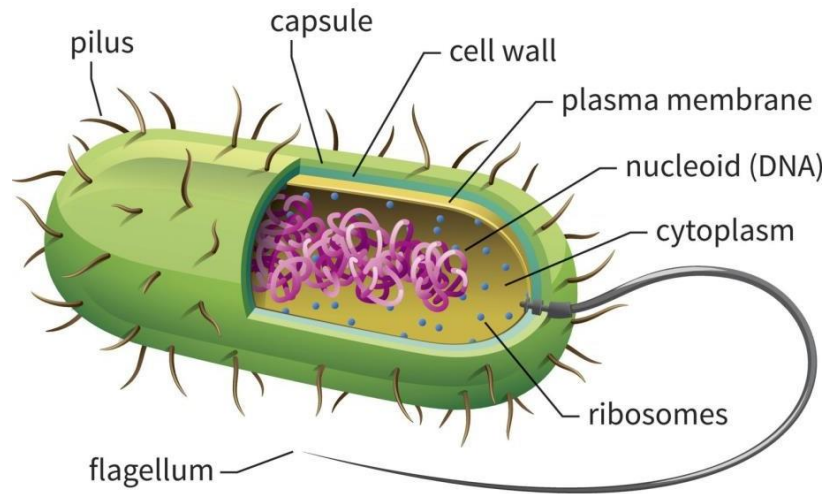
Prokaryotes

Organisms with prokaryotic cells are called prokaryotes (ex. Bacteria, blue green algae).

Single-celled microorganisms (unicellular).

These cells are smaller than eukaryotic cells and multiply in a faster rate.

They lack membrane bound nucleus.

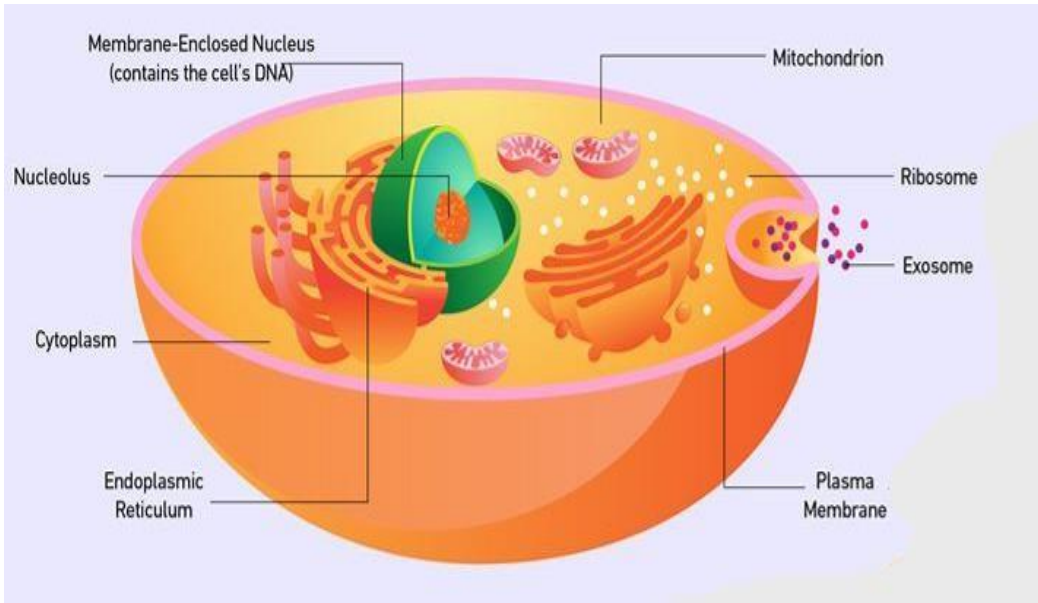


Prokaryotes

They are outermost covered with cell wall.

DNA is circular, without proteins.

The DNA is directly attached to cytoplasm.



Eukaryotes

Organisms with Eukaryotic cells are called Eukaryotes.
(ex. Plants, fungi, animals).

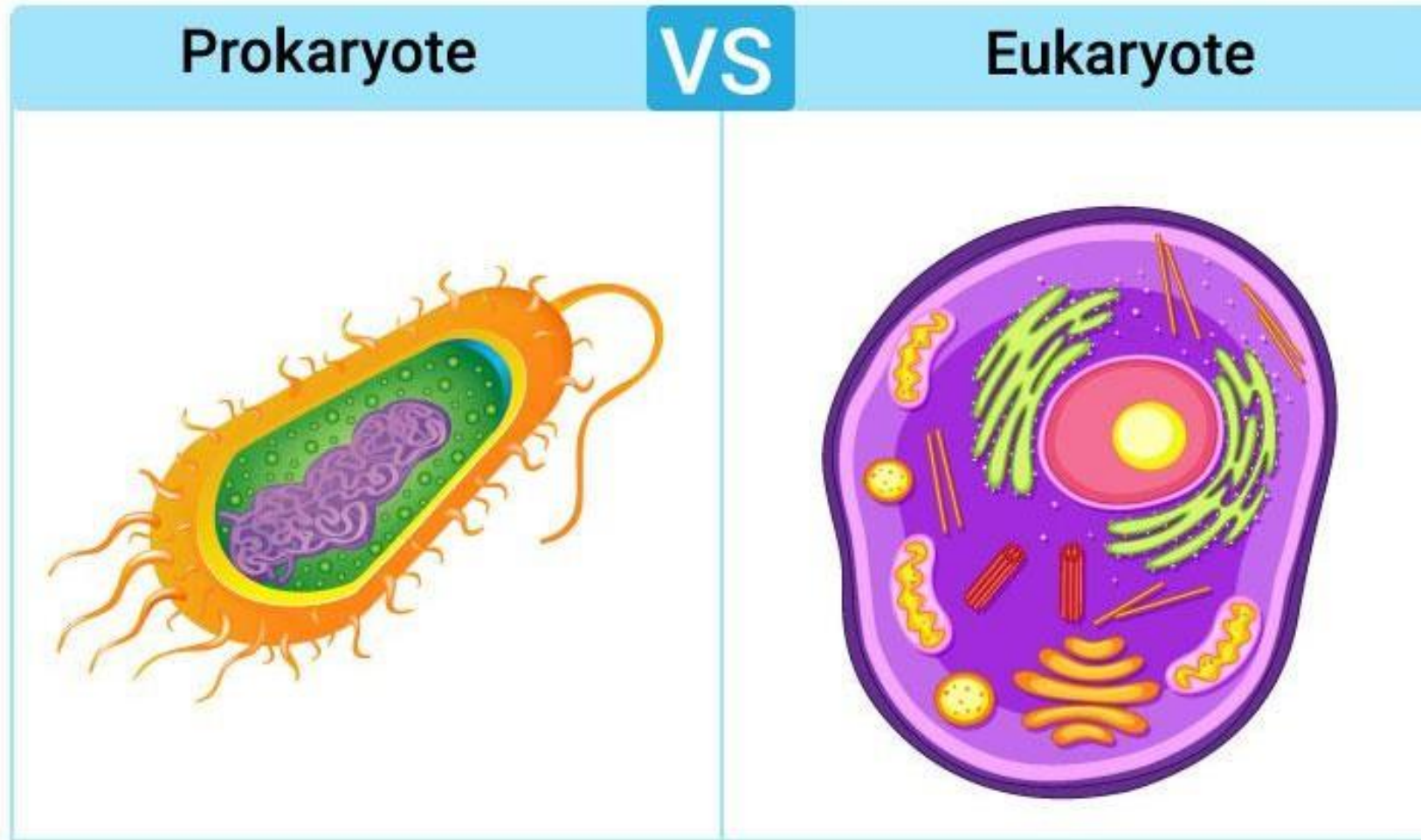
More complex structure than prokaryotes.

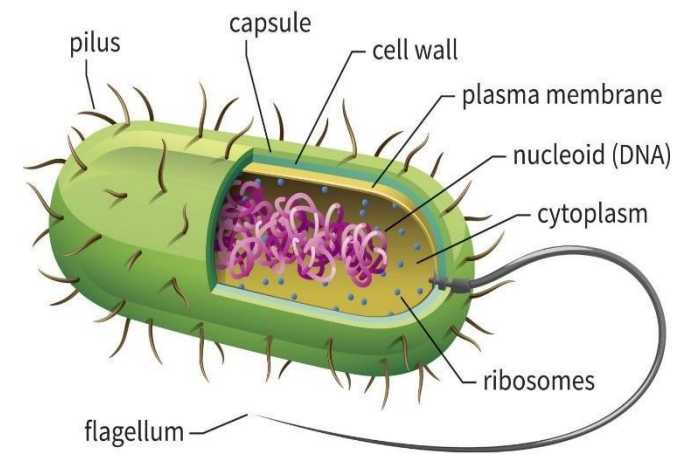
Larger than prokaryotes.

The nucleus is covered with nuclear membrane (true nucleus).

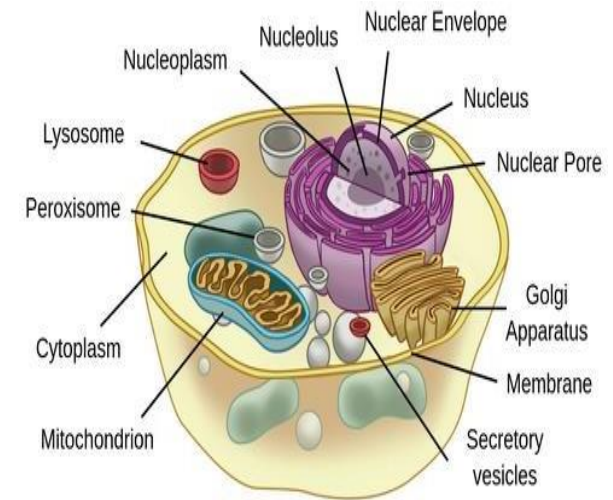
DNA is linear and associated with proteins

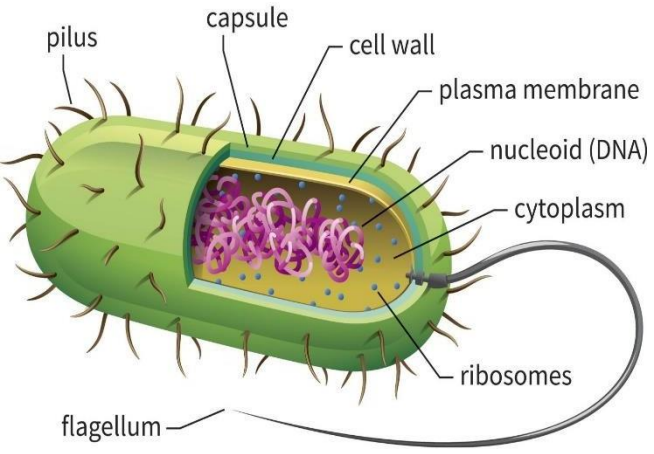
What are the differences ?



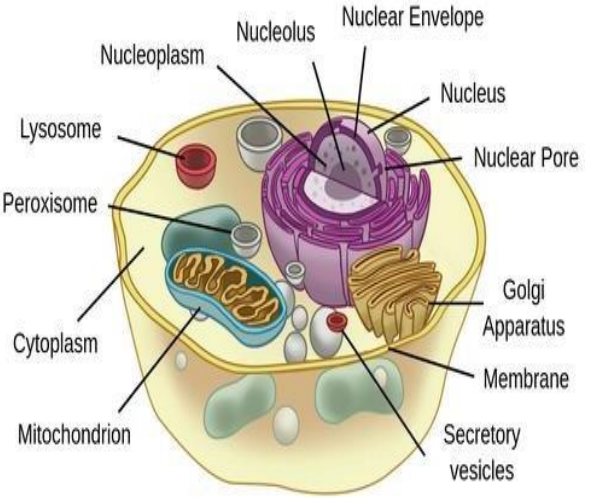


Prokaryotes	Eukaryotes
Organisms with prokaryotic cells	Organisms with Eukaryotic cells
(ex. Bacteria, blue green algae).	(ex. Plants, fungi, animals)
Single-celled microorganisms (unicellular)	May be unicellular or multicellular
Smaller in size than eukaryotes	Larger in size
More simple structure	More complex structure
Multiply in a faster rate	Multiply in a slower rate





Prokaryotes	Eukaryotes
Lack membrane bound nucleus (The DNA is directly attached to cytoplasm)	Have membrane bound nucleus (The DNA is separated from cytoplasm)
They are outermost covered with cell wall	Cell wall covers plant cells and fungi , but the cover of animal cell is called cell membrane
DNA is circular, without proteins	DNA is linear, with proteins



Bacteria

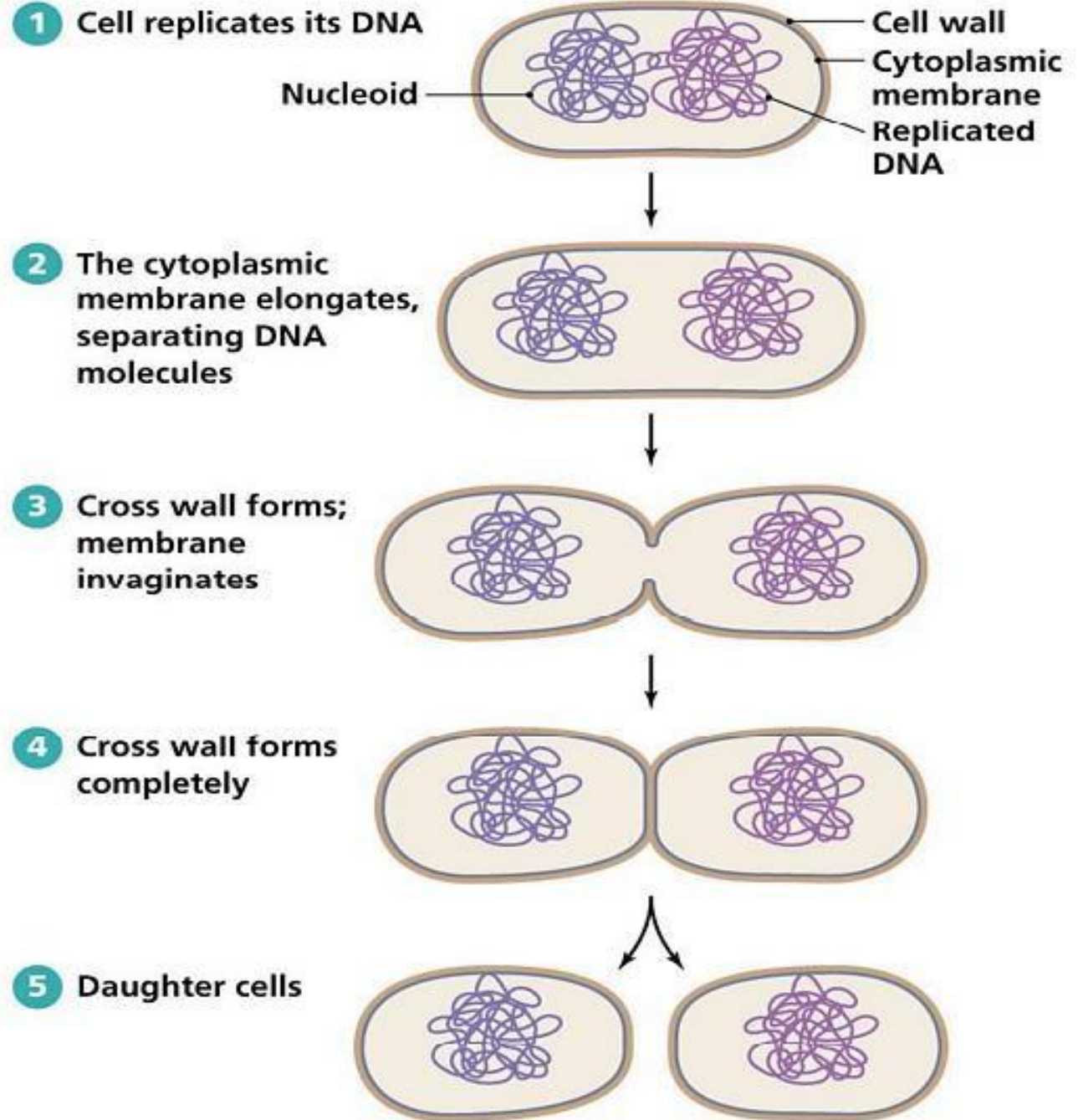
- Bacteria are the smallest and simplest microorganisms.
- They were firstly discovered by Antony van Leeuwenhoek.
- We will talk about binary fission and endospores of bacteria.



1- Binary Fission

- Binary fission is the simplest and the most common type of reproduction in bacteria.
- It occurs during favorable conditions.
- Binary fission involves the **division of a single cell**, which results in the formation of two identical cells.

Binary Fission Steps



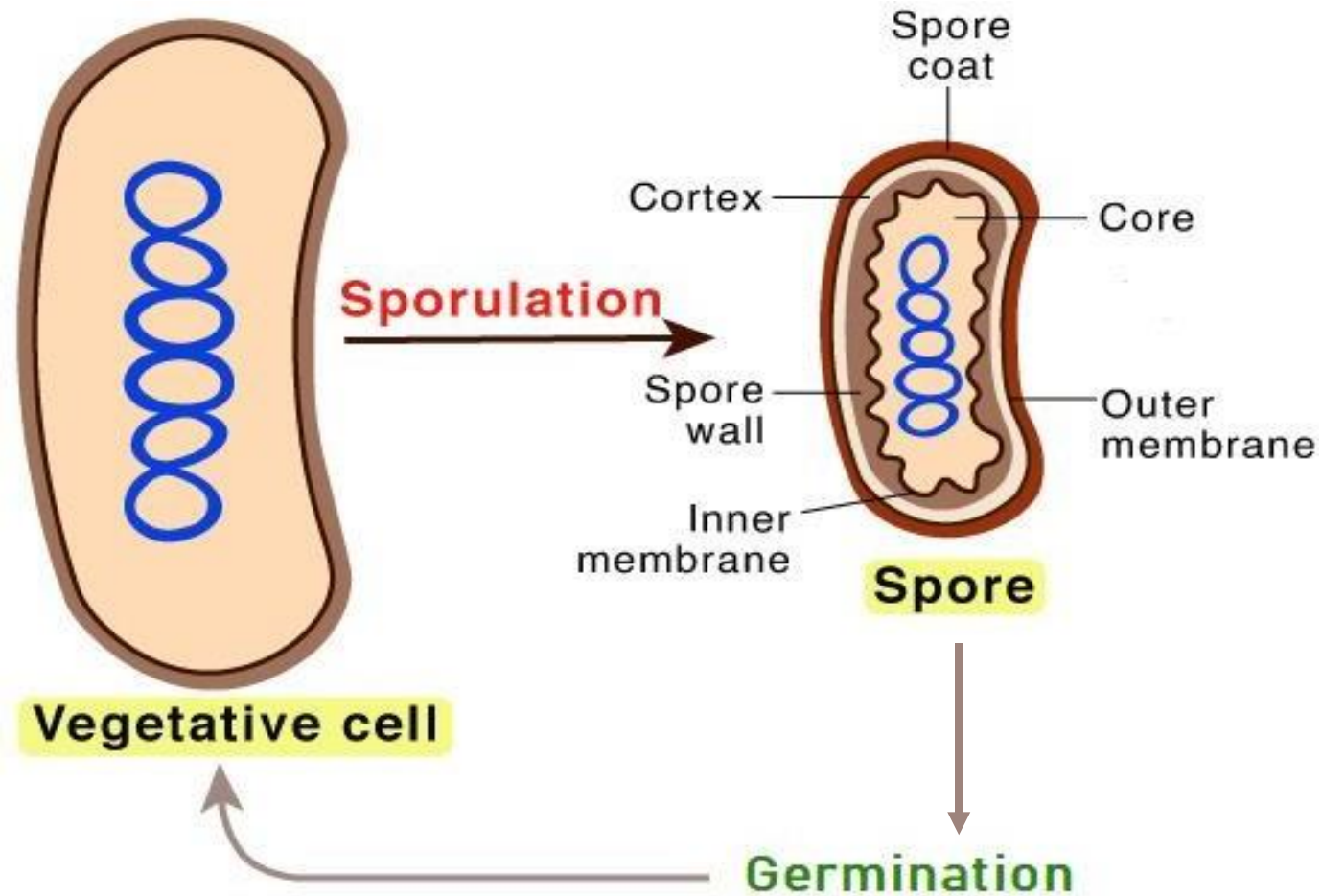
2- Endospores

- Endospores are formed during **unfavorable conditions** such as environmental stress.
- Bacterial cell is converted from vegetative form to spore.
- Endospore is resistant to temperature, radiation and chemical disinfectants (endospore is a resistant form for bacteria).
- Endospores can be found in soil, water, high temperature areas.
- They survive and live for a long time until the environment becomes good.
- This form is a static (not moving) bacteria.

2- Endospores

- **Sporulation** = Process of endospore formation from a vegetative cell.
- **Germination** = Return of an endospore to its vegetative form.

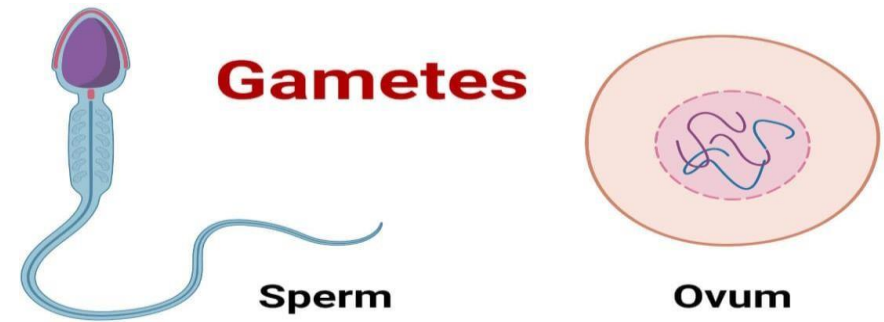
Sporulation and Germination



Sexual Reproduction in eukaryotes

Sexual Reproduction

- This is the way of reproduction of multicellular organisms.
- In this type of reproduction, male and female gametes (ova and sperms) are required.

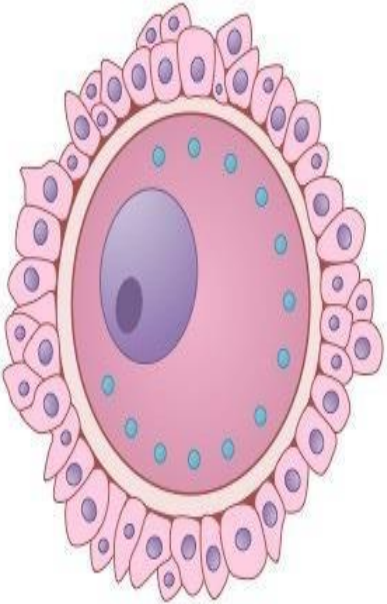


- The union of male and female gametes produces zygote which leads to the new individual.

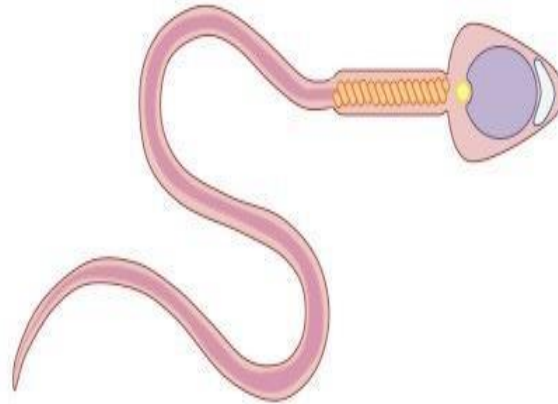


Germ cells

Human Egg (Ovum)



Human Sperm (Spermatozoa)



- One set of chromosomes.
- N.
- 23 chromosomes.

Somatic cells

Somatic Cells



Neuron



Epithelial Cells

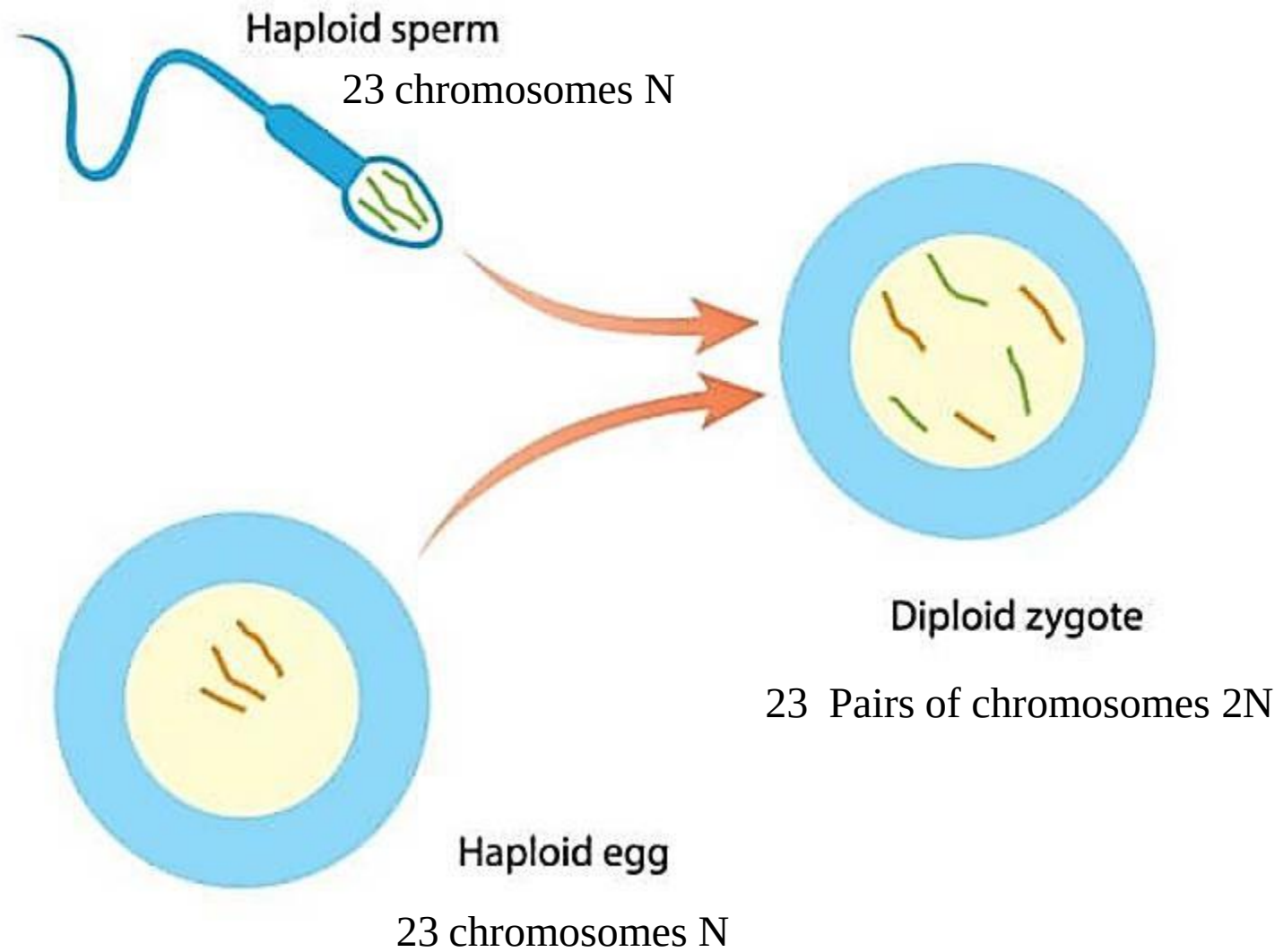
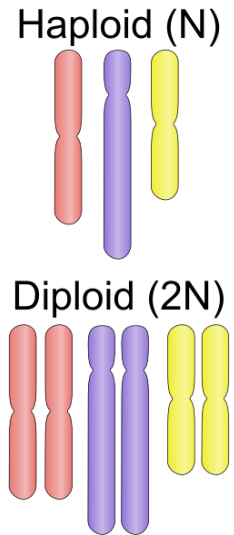


Blood Cells



Muscle Cells

- Two sets of chromosomes.
- 2N.
- 46 chromosomes or 23 pairs.



Questions

TRUE or FALSE?

- Binary fission occurs during favorable conditions ()
- Robert Hooke is the scientist who started cell theory ()
- Eukaryotes have more complex structure than prokaryotes ()
- Sporulation is the process of the formation of vegetative form ()
- In prokaryotes, the nucleus is covered with nuclear membrane ()
- Germ Cells are haploid cells (have two sets of chromosomes) ()
- Rudolph Virchow is the scientist who completed cell theory ()
- Animal cells have membrane bound nucleus, so animals are prokaryotes ()

GOOD
LUCK